Reformulating Imitation Learning as Return Distribution Matching

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Motivation

- Humans are *risk-sensitive* and *non-Markov*
- Existing Imitation Learning (IL) algorithms are *risk-neutral* and/or work with *Markov* policies

Contribution

- New **risk-sensitive** IL formulation
- *Provably efficient* algorithms with **non-Markov** policies when the reward is *known*
- Statistical insights when the reward is *unknown*

Return Distribution Matching

Formulation

When r^E is *known*:

$$\widehat{\pi} \in \arg\min_{\pi \in \Pi^{\mathrm{NM}}} \mathcal{W}\left(\eta_{r^E}^{\pi^E}, \eta_{r^E}^{\pi}\right)$$

When r^E is *unknown*:

$$\widehat{\pi} \in \arg\min_{\pi \in \Pi^{\mathrm{NM}}} \max_{r} \mathcal{W}\left(\eta_{r}^{\pi^E}, \eta_{r}^{\pi}\right)$$

Observations

- More expressive than vanilla IL $\rightarrow \left|J_{r^E}^{\pi^E} - J_{r^E}^{\widehat{\pi}}\right| \leq \mathcal{W}(\eta_{r^E}^{\pi^E}, \eta_{r^E}^{\widehat{\pi}})$
- Markov policies do not suffice:

  Proposition 1 *There exist an MDP $\mathcal{M}_{r^E}$ and $\pi^E \in \Pi^{NM}$ such that any Markov $\pi \in \Pi^{M}$ satisfies $\mathcal{W}(\eta_{r^E}^{\pi^E}, \eta_{r^E}^{\pi}) \geq 0.5$.*

- Stronger metrics (e.g., total variation) are sample inefficient:

  Theorem 2 *There exist an MDP $\mathcal{M}_{r^E}$ and $\pi^E \in \Pi^{NM}$ such that, even with $|\mathcal{D}^E| = (S-1)^{H-1} - 1$ trajectories, any algorithm satisfies $\mathbb{E}_{\mathcal{D}^E \sim \mathbb{P}^{\pi^E}} TV\left(\eta_{r^E}^{\pi^E}, \eta_{r^E}^{\widehat{\pi}}\right) > 0.1$.*

Imitation Learning

- r^E expert's reward
- $\pi^E \in \Pi^{\mathrm{NM}}$ (non-Markov) expert's policy
- $\mathcal{D}^E = \{(s_h, a_h)\} \sim \pi^E$ demonstrations
- Objective:

$$\widehat{\pi} \in \arg\min_{\pi \in \Pi^{\mathrm{NM}}} \left|J_{r^E}^{\pi^E} - J_{r^E}^{\pi}\right|$$

- When r^E is unknown:

$$\widehat{\pi} \in \arg\min_{\pi \in \Pi^{\mathrm{NM}}} \max_{r} \left|J_{r}^{\pi^E} - J_{r}^{\pi}\right|$$

- Equivalent to **occupancy measure matching**

$$\widehat{\pi} \in \arg\min_{\pi \in \Pi^{\mathrm{NM}}} \sum\nolimits_h \left\|d_h^{\pi^E} - d_h^{\pi}\right\|$$

- A minimum always exists in Π^{M}
- $\rightarrow$ **But** this does *not* control the return distribution

$$d^{\pi^E} \approx d^{\pi} \not\Rightarrow \eta_{r^E}^{\pi^E} \approx \eta_{r^E}^{\pi}$$

Unknown Reward Setting

Discretize any reward r:
$r_{\theta,h}(s,a) \coloneqq \arg\min_{x \in \{0,\theta,2\theta,\dots,\lfloor 1/\theta \rfloor \theta\}} |x - r_h(s,a)|$

Algorithm 1: RS-KT- variant

Input: $\mathcal{D}^E = \{(s_h^i, a_h^i)_h\}_i, p$
Define $\widehat{\eta}_r(g) \coloneqq \frac{1}{N}\sum_i \mathbb{1}\{G(\omega^i; r_\theta) = g\} \quad \forall r$
$\widehat{\pi} \in \arg\min_{\pi \in \Pi^{\mathrm{NM}}} \max_r \mathcal{W}\left(\eta_r^{\pi}, \widehat{\eta}_r\right)$ // oracle
Return $\widehat{\pi}$

Theorem 3 *Set $\theta = \epsilon/(2H)$. Then, with probability $1-\delta$, the policy $\widehat{\pi}$ output by the* oracle-based *Algorithm 1 satisfies $\max_r \mathcal{W}\left(\eta_r^{\pi^E}, \eta_r^{\widehat{\pi}}\right) \leq 2\epsilon$, with a number of samples:*

$$N \leq \widetilde{\mathcal{O}}\left(\frac{SAH^3}{\epsilon^2} \ln \frac{1}{\delta}\right)$$

Known Reward Setting

An Efficient Class of Non-Markov Policies

Adopt a **discretization approach**

- Given $\theta \in (0,1]$, build a discretized reward: $r_{\theta,h}^E(s,a) \coloneqq \arg\min_{x \in \{0,\theta,2\theta,\dots,\lfloor 1/\theta \rfloor \theta\}} |x - r_h^E(s,a)|$
- Markov policies in the return-augmented space: $\Pi(r_\theta^E) \coloneqq \left\{\pi \in \Pi^{\mathrm{NM}} \,\middle|\, \pi(a|s,\omega) = f(s, G(\omega; r_\theta^E), a, h)\right\}$

Lemma 4 *For any MDP $\mathcal{M}_{r^E}$ and $\pi^E \in \Pi^{NM}$, the set $\Pi(r_\theta^E)$ contains at least a π such that: $\mathcal{W}\left(\eta_{r^E}^{\pi^E}, \eta_{r^E}^{\pi}\right) \leq H\theta$.*

Algorithms

Algorithm 2: RS-BC

Input: $\mathcal{D}^E = \{(s_h^i, a_h^i)_h\}_i, r^E, \theta$
$N_h(s,g,a) \leftarrow \sum_i \mathbb{1}\{s_h^i = s, a_h^i = a, G(\omega_h^i; r_\theta^E) = g\}$
$\widehat{\pi}(a|s,\omega) \leftarrow \begin{cases} \frac{N_h(s, G(\omega; r_\theta^E), a)}{\sum_{a'} N_h(s, G(\omega; r_\theta^E), a')} & \text{if den} > 0 \\ \frac{1}{A} & \text{otherwise} \end{cases}$
Return $\widehat{\pi}$

Theorem 5 *Set $\theta = \epsilon/(4H)$. Then, with probability $1-\delta$, the policy $\widehat{\pi}$ output by Algorithm 2 satisfies $\mathcal{W}(\eta_{r^E}^{\pi^E}, \eta_{r^E}^{\widehat{\pi}}) \leq \epsilon$, with a number of samples:*

$$N \leq \widetilde{\mathcal{O}}\left(\frac{SH^6 \ln \frac{1}{\delta}}{\epsilon^3}\left(A + \ln \frac{1}{\delta}\right)\right)$$

Algorithm 3: RS-KT

Input: $\mathcal{D}^E = \{(s_h^i, a_h^i)_h\}_i, r^E, \theta, p$
$\widehat{\eta}(g) \leftarrow \frac{1}{N}\sum_{i \in [\![N]\!]} \mathbb{1}\{G(\omega_h^i; r_\theta^E) = g\}$
$\widehat{\pi} \in \arg\min_{\pi \in \Pi(r_\theta^E)} \mathcal{W}\left(\eta_{r_\theta^E}^{\pi}, \widehat{\eta}\right)$ // LP
Return $\widehat{\pi}$

Theorem 6 *Set $\theta = \epsilon/(7H)$ and assume that the optimization problem is solved exactly. Then, with probability $1-\delta$, the policy $\widehat{\pi}$ output by Algorithm 3 satisfies $\mathcal{W}(\eta_{r^E}^{\pi^E}, \eta_{r^E}^{\widehat{\pi}}) \leq \epsilon$, with a number of samples:*

$$N \leq \mathcal{O}\left(\frac{H^2}{\epsilon^2} \ln \frac{1}{\delta}\right)$$